Nithin Gopalakrishnan Nair

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Research Focus

Large-scale model training, generative modeling, multimodal learning, synthetic data, and 3D understanding.

Research Experience

Apple – Research Engineer, Multimodal Agents Jun 2025 – Present

- Designed an **auto-improvement pipeline** to improve the performance of long-horizon agents on **human interaction understanding** tasks.
- Owned **agent development infrastructure**, integrating simulation and evaluation workflows.
- Defined evaluation protocols and scoring rubrics to assess the quality and behavior of multimodal agents and guide model improvements.

Google – Google Beam Dec 2024 – Mar 2025

- Designed a novel transformer architecture for scaling novel view synthesis using diffusion-generated views (ICCV 2025; patent application filed).
- Achieved state-of-the-art novel view synthesis on Re10K, ACID, and DL3DV, and improved cross-dataset generalization using synthetic training data.

NVIDIA Research Apr 2024 – Dec 2024

- Built a zero-shot pipeline to segment **3D mesh parts** and estimate materials and physical properties using multi-view foundation-model features and multimodal reasoning (U.S. Patent Application 19/024,394).

Adobe Inc. May 2023 – Aug 2023

- Developed **Control2Control** for training-free image editing from sketches, poses, and layouts.
- Designed a novel attention transfer method for identity-preserving editing using human-drawn control signals.

Mitsubishi Electric Research Labs (MERL) May 2022 – Aug 2022

- Developed **Steered Diffusion**, an energy-based approach for zero-shot conditioning of diffusion models through inference-time guidance (ICCV 2023).

Research Highlights

Generative Model Training and Synthetic Data 2025

- Developed **GenDeg**, an approach for improving the image reconstruction performance of diffusion models on out-of-distribution data through augmentation with synthetic data (CVPR 2025).
- Discovered the connection between autoregressive generation and discrete diffusion in **Scale-Wise VAR** (WACV 2027; accepted). Based on this connection, proposed changes to improve VAR's scalability and generation performance.

Diffusion Steering and Compositional Generation 2022 – 2026

- Introduced **Unite and Conquer**, a generalized product-of-experts sampler that composes independently trained diffusion models without jointly labeled multimodal training data (CVPR 2023).
- Proposed **MaxFusion**, which composes off-the-shelf diffusion control modules through variance-aware feature fusion for multiple spatial conditions without retraining (ECCV 2024).
- Developed **D3SG**, which adaptively selects the guidance strength at each timestep for optimal diffusion guidance under linear and nonlinear conditions. *Under review.*

IARPA BRIAR – Unconstrained Face Recognition 2021 – 2025

- Developed a novel strategy to scale up training of transformer-based face-recognition models for extremely degraded images, achieving state-of-the-art results on five benchmarks [81.38% Rank-1 on IARPA BRIAR dataset] (FG 2025).
- Introduced a degradation-aware approach to recognizing faces in degraded images through large-scale pretraining and degradation-aware post-training. Achieved state-of-the-art results [76.66% Rank-1 on TinyFace dataset], (WACV 2025 Oral).

Selected Publications

18 peer-reviewed publications; full list on Google Scholar.

1. **Scaling Transformer-Based Novel View Synthesis Models with Token Disentanglement and Synthetic Data** 2025
   Nithin Gopalakrishnan Nair*, Srinivas Kaza*, Xuan Luo, Vishal M. Patel, Stephen Lombardi, Jungyeon Park
   IEEE/CVF International Conference on Computer Vision (ICCV)
2. **GenDeg: Diffusion-Based Degradation Synthesis for Generalizable All-in-One Image Restoration** 2025
   Sudarshan Rajagopalan, **Nithin Gopalakrishnan Nair**, Jay N. Paranjape, Vishal M. Patel
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
3. **MaxFusion: Plug&Play Multi-Modal Generation in Text-to-Image Diffusion Models** 2024
   Nithin Gopalakrishnan Nair, Jeya Maria Jose Valanarasu, Vishal M. Patel
   European Conference on Computer Vision (ECCV)
4. **CrowdDiff: Multi-hypothesis Crowd Density Estimation using Diffusion Models** 2024
   Yasiru Ranasinghe, **Nithin Gopalakrishnan Nair**, Wele Gedara Chaminda Bandara, Vishal M. Patel
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
5. **Steered Diffusion: A Generalized Framework for Plug-and-Play Conditional Image Synthesis** 2023
   Nithin Gopalakrishnan Nair, Anoop Cherian, Suhas Lohit, Ye Wang, Toshiaki Koike-Akino, Vishal M. Patel, Tim K. Marks
   IEEE/CVF International Conference on Computer Vision (ICCV)
6. **Unite and Conquer: Plug&Play Multi-Modal Synthesis Using Diffusion Models** 2023
   Nithin Gopalakrishnan Nair, Wele Gedara Chaminda Bandara, Vishal M. Patel
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
7. **PETALface: Parameter Efficient Transfer Learning for Low-Resolution Face Recognition** 2025
   Kartik Narayan, **Nithin Gopalakrishnan Nair**, Jennifer Xu, Rama Chellappa, Vishal M. Patel
   IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), ***Oral***

* Equal contribution.

Education

Johns Hopkins University, Baltimore, USA 2021 – 2025

- Ph.D. in Electrical and Computer Engineering, advised by Prof. Vishal M. Patel; GPA: *3.96/4.0*.

Indian Institute of Technology Madras, Chennai, India 2015 – 2020

- B.Tech. and M.Tech. Dual Degree in Electrical Engineering; CGPA: *9.0/10.0*.

Technical Skills

Languages: Python, C, C++, MATLAB
Frameworks: PyTorch, JAX
Generative Models: Diffusion models, autoregressive models, VAEs, GANs

Mentoring

Mentored five student researchers at Johns Hopkins University (2022–2025).

Academic Service

Reviewer: ICLR (2024–2026), NeurIPS (2024–2025), CVPR (2021, 2023, 2025), ICCV (2021, 2023, 2025), ECCV (2022, 2024); IEEE TPAMI, TIP, and TMI.